FMTH0306-1.0

Course Plan

Semester: V	Year: 2026-27
Course Title: Deep Learning	Course Code: 26ECAC302
Total Contact Hours: 50	Duration of ESA: 3Hrs
ISA Marks: 50	ESA Marks: 50
Lesson Plan Authors: Umadevi F M, Sneha Varur	Date: 12-08-2026
Checked By: Dr. Uday Kulkarni	Date: 10-09-2026

Prerequisites: Linear Algebra (24EMAB103), Multivariate Calculus (25EMAB224), Probability and Statistics (24EMAB104) and Machine Learning (25ECSC212).

Course Overview:

This course provides a concept-driven and application-oriented study of Deep Learning, progressing from neural network foundations, optimisation, and regularisation to CNNs, sequential models, attention and Transformers, self-supervised and generative learning, GNNs, and multimodal systems. Emphasis is placed on understanding the inductive bias, suitability, and limitations of different architectures, enabling students to select appropriate models based on the data structure and problem requirements. Theory is reinforced through hands-on laboratory experiments involving implementation, comparison, evaluation and failure analysis, culminating in an open-ended course project and the study of model evaluation, explainability, efficiency and responsible deployment.

Course Outcomes (CO'S):

At the end of the Course, the student should be able to:

1. **Explain** the fundamental concepts and learning mechanisms of deep learning.
2. **Apply** suitable architectures to learn spatial and sequential representations from data.
3. **Develop** models for contextual language understanding and representation learning from unlabelled data.
4. **Implement** models for generative, relational, and multimodal learning problems.
5. **Evaluate** learning models based on performance, generalisation, robustness, and computational efficiency.

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)

Course Title: Deep Learning	Semester: V
Course Code: 26ECAC302	Year:2026-27

Course Outcomes (COs) / Program Outcomes (POs)	Program Outcomes-PO's											PSO	
	1	2	3	4	5	6	7	8	9	10	11	1	2
1. Explain the fundamental concepts and learning mechanisms of deep learning.	H												
2. Apply suitable architectures to learn spatial and sequential representations from data.	H	H										H	
3. Develop models for contextual language understanding and representation learning from unlabelled data.	H		H									H	
4. Implement models for generative, relational, and multimodal learning problems.	H	H										M	
5. Evaluate learning models based on performance, generalization, robustness, and computational efficiency.		H	H									H	

Degree of compliance **L**: Low **M**: Medium **H**: High

Competency addressed in the Course, corresponding Performance Indicators (PI) and Course Specific Performance Indicators (CSPI)

Competency	Performance Indicator (PI)	Course Specific Performance Indicators (CSPI)
1.4: Demonstrate competency in Computer Science and Engineering	PI 1.4.4: Apply AI/ML techniques to design and develop intelligent, scalable systems.	1.4.4.1: Apply fundamental deep learning concepts, including artificial neural networks, activation functions, loss functions, and optimization techniques, to solve computational problems.
		1.4.4.2: Design and develop deep neural network architectures appropriate for classification, regression, and pattern recognition tasks.
		1.4.4.3: Implement convolutional neural networks for feature extraction and image classification applications.
		1.4.4.4: Apply recurrent neural networks and related architectures for sequential and time-series data analysis.
		1.4.4.5: Analyse and apply appropriate regularization, normalization, optimization, and hyperparameter tuning techniques to improve deep learning model performance.
		1.4.4.6: Design and implement transfer learning-based deep learning solutions using suitable pre-trained models for real-world applications.
		1.4.4.7: Evaluate and interpret deep learning models using appropriate performance metrics and error analysis techniques.
2.1: Demonstrate an ability to identify and define complex engineering problems by understanding real-world contexts and constraints.	PI 2.1.2: Identify critical systems/sub systems, variables, and constraints that characterize the problem environment by analysing practical contexts and real-world limitations.	2.1.2.1: Identify the key components, inputs, outputs, and variables involved in a deep learning problem based on the given application context.
		2.1.2.2: Analyse datasets and problem characteristics to identify relevant features, patterns, and relationships for deep learning-based solutions.
		2.1.2.3: Identify practical constraints such as data availability, quality, computational resources, model complexity, and scalability while formulating deep learning problems.
		2.1.2.4: Analyse real-world application requirements and select appropriate deep learning approaches based on problem characteristics and constraints.
Competency 3.3: Demonstrate an ability to select an optimal design solution considering cost-effectiveness and environmental impact.	PI 3.3.2: Build models/prototypes to develop chosen design solutions.	3.3.2.1: Design and develop deep learning models/prototypes to address identified computational and real-world problems.

		3.3.2.2: Implement and train appropriate deep learning architectures using suitable datasets, frameworks, and computational resources.
		3.3.2.3: Experiment with alternative model architectures, hyperparameters, and optimization strategies to develop an effective deep learning solution.
		3.3.2.4: Analyse the performance, computational requirements, scalability, and practical feasibility of developed deep learning models/prototypes to select an appropriate design solution.
PSO-2 (PO13) – Competency 1: Develop AI-based models to pre-process, analyse, and extract insights from structured and unstructured data.	PI 13.1.1: Build AI/ML models for data engineering tasks.	13.1.1.1: Pre-process and prepare structured and unstructured datasets for training deep learning models.
		13.1.1.2: Develop deep learning models using appropriate architectures for extracting meaningful features from structured, image, text, and sequential data.
		13.1.1.3: Implement and train deep learning models using suitable frameworks and computational tools.
		13.1.1.4: Apply data augmentation and feature representation techniques to improve the generalization capability of deep learning models.
PSO-2 (PO13) – Competency 1: Develop AI-based models to pre-process, analyse, and extract insights from structured and unstructured data.	PI 13.1.2: Evaluate efficiency, scalability, and accuracy of AI-based solutions.	13.1.2.1: Evaluate deep learning models using appropriate accuracy, precision, recall, F1-score, loss, and other task-specific performance measures.
		13.1.2.2: Compare alternative deep learning architectures and optimization strategies based on predictive performance and computational requirements.
		13.1.2.3: Analyse overfitting, underfitting, convergence, and generalization behaviour of deep learning models.
		13.1.2.4: Optimize deep learning models to improve computational efficiency, scalability, and predictive performance.
PSO-2 (PO13) – Competency 2: Develop AI-driven solutions to optimize, secure, and	PI 13.2.1: Develop AI-based optimized solutions for system and network management in diverse	13.2.1.1: Apply deep learning techniques to develop intelligent solutions for domain-specific problems in areas such as healthcare, agriculture, industry, and other application domains.

manage systems and networks.	application domains (e.g., agriculture, healthcare, industry).	
		13.2.1.2: Select and adapt suitable deep learning architectures based on the characteristics and requirements of a given application domain.
		13.2.1.3: Develop and evaluate deep learning-based solutions that address real-world prediction, classification, detection, or decision-making problems.
		13.2.1.4: Analyse the practical limitations, computational requirements, and deployment considerations of deep learning solutions in real-world environments.

Course Content

Course Code: 26ECAC302	Course Title: Deep Learning	
L-T-P: 4-0-0	Credits: 4	Tutorial/Practical: --
ISA Marks: 50	ESA Marks: 50	Total Marks: 100
Teaching Hrs: 50	Tutorial Hrs: --	Exam Duration: 3 hrs

Content	Hrs.
Unit – 1	
Chapter 1: Foundations and Representation Learning The Architecture Landscape: Deep Learning, The Representation Problem, The Artificial Neuron and the Multilayer Perceptron, Activation Functions, Loss Functions as Task Objectives, How Networks Learn: Forward and Backward Propagation, Optimisation: Making Gradient Descent Work, Generalisation, Overfitting, and Regularisation, Reading the Training Signal: Debugging Deep Networks, When to Use Deep Learning — and When Not To, Chapter Resolution and Bridge.	**03 hrs**
Chapter 2 Learning from Images — CNNs and Visual Understanding Image Analysis and Chapter Overview, Image Data as a Deep Learning Context, Traditional Image Processing: Fixed Operators Before Learned Features, Hand-Crafted Feature Extraction: Turning Images into Measurements, The Traditional Computer-Vision and Machine-Learning Pipeline, Traditional Pipeline Limitations, Representation Learning: From Human-Designed Features to Learned Features, The Inductive Bias for Image Data, Fully Connected Networks and Image Data, The Convolution Operation, Pooling and Spatial Down-sampling, CNN Architecture: Assembling the Components, Architecture Evolution as Problem–Response, Visual Tasks Beyond Classification, Transfer Learning in Computer Vision, CNN Limitations, Chapter Resolution and Bridge	**09 hrs**
Chapter 3 Learning from Sequences — RNNs, LSTMs, and Temporal Patterns The Inductive Bias for Sequential Data, Time-Series Data as a Deep Learning Context, Feedforward Networks and Sequential Data, The Recurrent Neural Network, Long Short-Term Memory, Gated Recurrent Units, 1D Convolutional Networks for Sequences, Sequence Architecture Patterns and Application Contexts, Sequence Model Limitations, Chapter Resolution and Bridge	**06 hrs**
Unit – 2	
Chapter 4 Learning from Text — Attention, Transformers, and Language Models Natural Language Processing: Foundations of NLP, Text Representation, Capturing Word Relationships, Word Embeddings, Classical NLP Methods and Challenges, The NLP Pipeline, The LLM Era, The Inductive Bias for Text Data, Text as Data, Sequential Models and Text Data, The Attention Mechanism, Self-Attention and Multi-Head Attention, The Transformer Architecture, Pre-training and the Foundation Model Paradigm, Fine-Tuning, Adaptation, and Efficient Transfer, Transformers Beyond Text, Hallucination, Grounding, and Responsible Use, NLP Application Contexts, Chapter Resolution and Bridge	**08 hrs**

Chapter 5. Learning Without Labels — Autoencoders and Self-Supervised Learning The Inductive Bias for Unlabelled Data, Limits of Supervised Learning, Autoencoder Architecture, Latent Space and Applications, Self-Supervised Learning, Application Contexts, Limitations, Chapter Resolution and Bridge.	**07 hrs**
Chapter 6. Generative Deep Learning — From Prediction to Synthesis The Inductive Bias for Generation, Generative Deep Learning, Variational Autoencoders, Generative Adversarial Networks, Diffusion Models, Application Contexts and Responsible Use	**07 hrs**
Unit – 3	
Chapter 7 Relational and Multimodal Data — GNNs and Beyond The Inductive Bias for Relational Data, Graph-Structured Data, Graph Neural Networks, Multimodal Deep Learning, Deep Reinforcement Learning: A Conceptual Survey, Reinforcement Learning in Agentic AI Applications, Chapter Resolution and Bridge	**10hrs**

Text Books

1. Ian Goodfellow and Yoshua Bengio and Aaron Courville: Deep Learning, MIT Press, 2016.

References

1. Luca Pietro Giovanni Antiga, Thomas Viehmann, Eli Stevens, Deep Learning with PyTorch Manning Publications, 2020.

Evaluation Scheme

ISA Scheme

Assessment	QP Max Marks	Weightage in Marks
In Semester Assessment -1 (ISA1)	40	20
In Semester Assessment -2 (ISA2)	40	20
Course Project	20	10
TOTAL	100	50

Course Unitisation for ISA and ESA

Topics / Chapters	Teaching Hours	No. of Questions in ISA-1	No. of Questions in ISA-2	No. of Questions in Activity	No. of Questions in ESA
Unit I					
1. Foundations and Representation Learning	3	0.5		–	0.5
2. Learning from Images — CNNs and Visual Understanding	9	1.5		–	1.5
3. Learning from Sequences — RNNs, LSTMs, and Temporal Patterns	6	0.5		–	0.5
Unit II					
4. Learning from Text — Attention, Transformers, and Language Models	8	-	1	–	1
5. Learning Without Labels — Autoencoders and Self-Supervised Learning	7	-	1	–	1
6. Generative Deep Learning — From Prediction to Synthesis	7	-	1	–	1
Unit III					
7. Relational and Multimodal Data — GNNs and Beyond	10	-	–	–	2

Note

1. Each Question carries 20 marks and may have sub-questions.
2. Mixing of sub-questions from different chapters within a unit (only for Unit I and Unit II) is allowed in ISA I, II and ESA
3. Answer 5 full questions of 20 marks each (two full questions from Unit I, II and one full question from Unit III) out of 8 questions in ESA.

Course Assessment Plan

Course Title: Deep Learning					**Code:** 26ECAC302
Course outcomes (COs)	**Weightage in assessment**	**Assessment Methods**			
		ISA - I	**ISA – II**	**Course Activity**	**ESA**
1. Explain the fundamental concepts and learning mechanisms of deep learning.	15%	✓			✓
2. Apply suitable architectures to learn spatial and sequential representations from data.	25%	✓			✓
3. Develop models for contextual language understanding and representation learning from unlabelled data.	20%		✓		✓
4. Implement models for generative, relational, and multimodal learning problems.	20%		✓		✓
5. Evaluate learning models based on performance, generalization, robustness, and computational efficiency.	20%			✓	✓
Weightage:	100%	20%	20%	10%	50%

Date: 16-09-20206

HoD, CSE(AI)

Course Project Evaluation Rubrics

Parameter	Good (4M – 5M)	Average (2M-3M)	Poor (0M – 1M)	BL	CO	PI
Problem statement identification (Review-1)	Finalization of the problem statement and domain knowledge is well understood with an exhaustive literature survey related to the problem statement is performed.	Problem statement is identified, with poor domain understanding and proper literature survey not carried out.	Problem statement is identified and no domain understanding and no literature survey.	3	2	1.4.4 2.1.2
Parameter	**Good (4M – 5M)**	**Average (2M-3M)**	**Poor (0M – 1M)**	**BL**	**CO**	**PI**
Assessing the Implementation (Review-2)	Complete implementation of the course project along with accurate model performance results.	80% - 90% implementation of the course project along with moderate model performance results.	60%-70% implementation of the course project with poor model performance results.	4	4	3.3.2 13.1.1
Parameter	**Good (4M – 5M)**	**Average (2M-3M)**	**Poor (0M – 1M)**	**BL**	**CO**	**PI**
The demonstration of the course project (Review-3)	Model is completely working for all the test cases and with good training and testing performance.	Model is working for few test cases and with moderate training and testing performance.	Model is not working for test cases and with poor training and testing performance.	3	5	13.1.2

Chapter wise Plan

Course Code and Title: 26ECAC302 Deep Learning	
Chapter Number and Title: 1. Foundations and Representation Learning	Planned Hours:03

Learning Outcomes:

At the end of the topic the student should be able to:

Sr.No	TLO's	CO's	BL	CA Code
1.	Explain Deep Learning architectures, motivation, and representation learning.	2	L2	13.1
2.	Apply artificial neurons, MLPs, activation/loss functions, and propagation to understand how the model learns.	1	L3	1.4
3.	Apply and analyse optimisation techniques and training processes for effective network learning.	4	L3	13.1
4.	Analyse generalisation, overfitting, regularisation, and training issues using training signals.	3	L2	2.1
5.	Evaluate Deep Learning's applicability and limitations and integrate concepts into an end-to-end workflow.	5	L3	13.1

Lesson Schedule Class No. - Portion covered per hour	Hours
1. Architecture Landscape, Deep Learning Motivation & Representation Learning Artificial Neuron, MLP, Activation & Loss Functions	1
2. Forward/Backward Propagation & Optimisation Generalisation, Overfitting & Regularisation	1
3. Training-Signal Analysis, Network Debugging, Applicability & Limitations Chapter Integration, Resolution & Bridge to Advanced Architectures	1

Review Questions

Sr.No. - Questions	TLO	BL	PI Code
1. MedSense is developing a system to predict whether a patient is likely to experience a cardiac deterioration event within the next 7 days using: i. Continuous ECG recordings from a wearable device, ii. Patient's previous clinical measurements and hospital records, and iii. Cardiologist's latest written clinical assessment.	1	L3	1.4.4

<table>
<tr><td>(a) Identify the three input modalities, classify each according to the Architecture Decision Map, and recommend an appropriate Deep Learning architecture family for processing each modality. Justify your choices.

(b) Formulate the problem as a supervised learning task. Specify the target variable, output type, suitable output activation function, and loss function. Explain why each is appropriate for the given prediction problem.</td><td></td><td></td><td></td></tr>
<tr><td>2. From Table below, the unregularised MLP (RMSE 241) is worse than the regularised MLP (RMSE 219), which is worse than gradient boosting (RMSE 198). Explain why each gap exists. What does the first reveal about regularisation? The second about the suitability of deep learning for this task?
<table>
<tr><th>Model</th><th>Details</th><th>Test RMSE (kg/acre)</th><th>Test MAE (kg/acre)</th><th>Training time</th></tr>
<tr><td>Linear regression</td><td>—</td><td>312</td><td>258</td><td>< 1 s</td></tr>
<tr><td>Decision tree</td><td>max_depth = 8</td><td>286</td><td>231</td><td>< 1 s</td></tr>
<tr><td>Gradient boosting</td><td>200 trees, depth 4</td><td>198</td><td>164</td><td>3 s</td></tr>
<tr><td>MLP (no regularisation)</td><td>7→64→32→1, ReLU</td><td>241</td><td>196</td><td>45 s</td></tr>
<tr><td>MLP (dropout + weight decay)</td><td>7→64→32→1, p=0.3, λ=10⁻³</td><td>219</td><td>178</td><td>45 s</td></tr>
<tr><td>MLP (larger, regularised)</td><td>7→128→64→32→1</td><td>214</td><td>174</td><td>80 s</td></tr>
</table></td><td>2</td><td>L3</td><td>2.1.2</td></tr>
<tr><td>3. A binary motor fault-detection MLP using flattened vibration spectrograms achieves 51.8% training and 51.1% validation accuracy, with losses near 0.69. Diagnose the primary failure mode (underfitting, overfitting, poor optimisation, or architectural choice) using the loss, accuracy, and generalisation gap. Also state whether the initial loss indicates random predictions.</td><td>4</td><td>L3</td><td>13.1.2</td></tr>
<tr><td>4. A smart manufacturing unit has 600 labelled thermal images of electrical panels (420 normal, 180 faulty) and requires an automated fault-detection model to run on an edge device with less than 150 ms inference time per image.

(a) Using a Deep Learning model-selection decision framework, evaluate the suitability of Deep Learning considering dataset size, class imbalance, computational constraints, and image characteristics. Recommend an appropriate training strategy and architecture family.

(b) Design an evaluation and deployment strategy. Justify suitable performance metrics beyond accuracy, and explain how you would address overfitting, class</td><td>4</td><td>L3</td><td>3.3.2</td></tr>
</table>

imbalance, and the inference-time constraint. Briefly state when a traditional ML model using hand-crafted features might be preferable			
5. A smart factory currently uses gradient boosting on engineered sensor features to predict machine remaining useful life (RUL). Under the existing setup, the neural network performs worse. What three specific changes to the data, model, or training setup could make the neural network competitive with gradient boosting? Explain how each change would improve the neural network's ability to learn complex patterns and shift the performance comparison.	5	L3	1.4.4

Chapter wise Plan

Course Code and Title: 26ECAC302 Deep Learning	
Chapter Number and Title: 2 and Learning from Images—CNNs and Visual Understanding	Planned Hours: 09

Learning Outcomes:

At the end of the topic the student should be able to:

Sr.No	TLO's	CO's	BL	CA Code
1	Explain the challenges involved in image analysis and the role of traditional computer vision techniques.	01	L2	1.4
2	Apply traditional image processing and hand-crafted feature extraction techniques for image analysis.	01	L3	1.4
3	Analyze the limitations of machine-learning methods and justify the need for representation learning.	02	L3	2.1
4	Explain the principles and key components of CNNs for hierarchical feature learning.	02	L2	13.1
5	Compare major CNN architectures based on their architectural innovations and applications.	03	L3	13.1
6	Apply fine-tuning for visual tasks and distinguish classification, detection, and segmentation.	05	L3	13.2

Lesson Schedule Class No. - Portion covered per hour	Hours
1. AgriVision task and learning path, Why image analysis is difficult, Traditional image processing	1
2. Hand-crafted features, Traditional ML pipeline, Why the traditional route plateaus	1
3. Representation learning, Why MLPs fail on images	1
4. Inductive bias for images, Convolution, Pooling and down sampling	1
5. Complete CNN architecture, Training and evaluation	1
6. Architecture evolution	1
7. Transfer learning	1
8. Visual task formulations	1
9. Failure modes, Resolution and bridge	1

Review Questions

Sr.No. - Questions	TLO	BL	PI Code
1.What are the major challenges in analyzing real-world images for computer vision applications? Explain any two with examples.	01	L2	1.4.4
2.An AgriVision image is captured at 4,032 × 3,024 pixels and resized to 224 × 224 pixels. Calculate the number of pixels before and after resizing, and explain why resizing is necessary for CNN-based image analysis.	02	L3	13.1.1.
3.Given the following 3 × 3 input image matrix: I = [10 10 10] [10 50 50] [10 50 50] Using the Sobel operators: G_x = [−1 0 1] [−2 0 2] [−1 0 1] G_y = [−1 −2 −1] [0 0 0] [1 2 1] Calculate the horizontal gradient G_x, vertical gradient G_y, and gradient magnitude at the centre pixel.Gradient magnitude: $\|G\| = \sqrt{(G_x^2 + G_y^2)}$	01	L3	1.4.4
4.An MLP receives a 224×224 RGB image as input and uses a hidden layer of 512 neurons. Explain how flattening the image affects the number of trainable parameters and why this makes MLPs inefficient for image classification.	03	L3	13.1.1
5.A CNN is trained to detect a disease pattern that may occur at different locations on a leaf. Explain how local receptive fields and weight sharing help the CNN detect the same pattern across different image locations, and why this is more efficient than an MLP.	04	L3	2.1.2

6.A CNN takes a 32 × 32 × 3 input image and applies a 3 × 3 convolution with 16 filters, stride 1, and padding 1. Calculate the output tensor dimensions and total number of trainable parameters, including one bias for each filter.	04	L3	13.1.1
7.A CNN feature map of size 28 × 28 × 128 is given as input to a Global Average Pooling (GAP) layer followed by a fully connected layer for 4-class classification. Determine the output size after GAP and the number of trainable parameters in the fully connected layer.	04	L3	13.2.1
8.Apply AlexNet Architecture for image classification task to classify 60 different categories of flowers draw the diagram and explain how each layer processes the image and justify how its major innovations contributed to improved classifications	05	L3	13.2.1
9.An input feature map of size 128x128x64 is processed by: (i) a regular 3x3 convolution with 128 filters and (ii) a depthwise 3x3 convolution followed by a 1x1 pointwise convolution with 128 filters. Calculate the parameters for both cases and the percentage reduction obtained using depthwise separable convolution. Ignore biases.	05	L3	13.2.1
10. A YOLO model detects three lesions of the same disease with confidence scores 0.92, 0.78, and 0.65. Two bounding boxes have IoU = 0.68, while the third has IoU = 0.20 with the highest-confidence box. (a) Which box is selected first during NMS? (b) Which box is suppressed using an IoU threshold of 0.5? (c) Explain how NMS helps produce the final detection result.	05	L3	13.2.1
11. A plant-disease detection system must identify the exact affected pixels of a leaf under varying disease shapes and sizes. Analyze how U-Net's encoder, decoder, and skip connections can be designed to preserve both high-level semantic information and fine spatial details for accurate segmentation.	06	L3	3.3.2

Chapter wise Plan

Course Code and Title: 26ECAC302 Deep Learning	
Chapter Number and Title: 3 Learning from Sequences — RNNs, LSTMs, and Temporal Patterns	Planned Hours:6

Learning Outcomes:

At the end of the topic the student should be able to:

Sr.No	TLO's	CO's	BL	CA Code
1	Explain the characteristics of sequential data and temporal dependencies	1	L2	1.4
2	Apply the principles of sequence models to process sequential	2	L3	2.1

	data.			
3	Analyse the suitability of sequential models for different sequence learning tasks.	2	L3	2.1
4	Compare the performance and limitations of sequence models.	5	L3	13.1

Lesson Schedule Class No. - Portion covered per hour	Hours
1. The Inductive Bias for Sequential Data, Time-Series Data as a Deep Learning Context	1
2. Why Feedforward Networks Struggle with Sequences: The Recurrent Neural Network.	1
3. Long Short-Term Memory.	1
4. Gated Recurrent Units.	1
5.1D Convolutional Networks for Sequences	1
6. Sequence Architecture Patterns and Application Contexts, When Sequence Models Fail.	1

Review Questions

Sr.No. - Questions	TLO	BL	PI Code
1. What is a Recurrent Neural Network (RNN), and how does it differ from feedforward neural networks?	2	L2	1.4.4
2. Discuss the challenges of training traditional RNNs, such as the vanishing gradient problem, and how they can affect the learning of long- range dependencies.	1	L2	1.4.4
3. A farmer's monitoring system stores insect counts from the previous 45 days as independent input features to an MLP. What are two limitations of this fixed-window representation compared to an LSTM?	3	L3	13.2.1
4. A hospital develops a system to monitor patients' heart rate continuously in real time. (a) Select the most suitable model among RNN, LSTM, and TCN. (b) Justify your selection based on sequence length and latency. (c) Identify one limitation of the chosen model.	6	L3	2.1.2
5. A deep RNN experiences the vanishing gradient problem during training. Explain how LSTM or GRU architectures address this issue and identify which architecture you would recommend for long	4	L3	2.1.2

sequences.			
6. A sequence of length T = 20 is processed using: • A TCN with 3 layers, kernel size 3, and 32 channels. • An LSTM with hidden size H = 32. Apply the computational complexity formulas to determine which model requires fewer operations for one forward pass. Justify your answer.	5	L3	13.1.1

Chapter wise Plan

Course Code and Title: 26ECAC302 Deep Learning	
Chapter Number and Title: 4. Learning from Text — Attention, Transformers, and Language Models	Planned Hours: 08

Learning Outcomes:

At the end of the topic the student should be able to:

Sr.No	TLO's	CO's	BL	CA Code
1.	Explain text representation, its inductive bias, and limitations of sequential models.	2	L2	1.4
2.	Apply and analyse attention mechanisms for contextual text representation.	3	L3	1.4
3.	Explain and analyse the Transformer architecture and its role in modern language models.	3	L3	2.1
4.	Evaluate pre-training and efficient transfer techniques for NLP tasks.	3	L3	13.1
5.	Assess Transformers across domains, including grounding, hallucination, and responsible use.	3	L3	13.2

Lesson Schedule Class No. - Portion covered per hour	Hours
1. Natural Language Processing-Foundations of NLP, Text Representation, Capturing word relationships,	1
2. Word Embeddings, Classical NLP methods and challenges, The NLP Pipeline, The LLM Era	1
3.Text as Data, Inductive Bias & Limitations of Sequential Models	1
4.Attention, Self-Attention & Multi-Head Attention	1
5.Transformer Architecture	1

6.Pre-training, Foundation Models, Fine-Tuning & Efficient Transfer	1
7.Transformers Beyond Text, Hallucination, Grounding & Responsible Use	1
8.NLP Applications & Chapter Resolution	1

Review Questions

Sr.No. - Questions	TLO	BL	PI Code
1. A farmer sends the following message to an agricultural advisory system: “The tomato plants that were treated with fertilizer after continuous rainfall have developed yellow patches.” The words “tomato plants” and “developed yellow patches” are separated by several other words. If you are designing a deep learning model to understand this message, explain why a Transformer would be more suitable than an MLP or CNN. Relate your answer to context dependency, sequential nature, and long-range dependency in text.	1	L3	2.1.2
2. An agricultural monitoring system receives a 500-word report describing the history of a crop from planting to disease detection. The important information about the crop appears near the beginning, while the disease information appears near the end.If an RNN is used to process this report, identify the major problems that may occur. Explain how sequential processing, long-range dependency, vanishing gradients, and limited parallelization can affect the system's performance.	1	L3	1.4.4
3. Consider the sentence: “The farmer who visited the research center last month finally planted tomatoes.” An NLP system is processing the word “planted.” Using the concept of the Attention Mechanism, explain how the model can directly identify “farmer” as an important word without passing information sequentially through all the intermediate words. Explain the roles of Query, Key, Value, Softmax, and weighted sum in this process	2	L3	1.4.4
4. An agricultural advisory system receives the sentence:“The experienced farmer carefully sprayed the pesticide on the infected crop yesterday.” The system needs to understand several relationships simultaneously: farmer → sprayed, pesticide → sprayed, infected → crop, and yesterday → sprayed. Would single self-attention or multi-head attention be more appropriate for this task? Justify your choice and explain how different attention heads can capture different semantic or syntactic relationships simultaneously	2	L3	1.4.4
5.An organization wants to process millions of agricultural reports using GPUs. The existing RNN-based system processes one word at a time and takes a very long time	3	L3	3.3.2

to train. Design a Transformer-based solution for this problem. Explain how the Transformer encoder, self-attention, positional encoding, and parallel processing can overcome the limitations of the existing RNN system.			
6.An agricultural organization wants to develop three separate systems: • System A: Classifies farmer complaints into categories such as disease, irrigation, fertilizer, and pest. • System B: Acts as a conversational agricultural chatbot. • System C: Converts an English agricultural advisory into Kannada. Choose the most suitable model among BERT, GPT, and T5 for each system. Justify your choices based on their architecture and capabilities.	4	L3	13.1.1
7.A law firm has a pretrained language model but only has a small collection of legal documents and limited GPU resources. Full fine-tuning would require updating millions of parameters and may cause catastrophic forgetting.Which adaptation strategy—Feature Extraction, Full Fine-Tuning, or LoRA—would you recommend? Explain your choice and describe how LoRA can adapt the model while keeping most pretrained parameters frozen.	4	L3	13.1.1
8.A computer vision research team has approximately 8,000 labelled object images. They want to classify objects from images and are considering ResNet-50, a pretrained ViT, and CLIP. Which model would you recommend for this dataset? Compare the three approaches with respect to data requirements, pretraining, spatial inductive bias, zero-shot classification, and domain-specific fine-tuning. Also explain when ViT or CLIP could become more competitive.	5	L3	2.1.2
9.A farmer asks an AI system:“What treatment should be given for early-stage powdery mildew on flowering tomato plants?”The standalone LLM generates a confident treatment recommendation, but it cannot provide any supporting source.Design a RAG-based solution for this problem. Explain the complete workflow from query embedding → vector database search → top-k document retrieval → prompt construction → LLM generation → grounded response with citations. Explain how this approach reduces hallucination.	5	L3	3.3.2
10.A Transformer has 8 weight matrices, each of size 2048 × 2048. LoRA is applied separately with ranks r = 4, 8, and 16. Calculate for each rank: 1. Number of trainable parameters. 2. Percentage of parameters compared with full fine-tuning. 3. How many more parameters are trained when increasing the rank from 4 → 8 and 8 → 16.	5	L3	13.1.2

11.An agricultural organization wants to build a complete AI-based farmer advisory system. A farmer enters: “How should early-stage powdery mildew on flowering tomato plants be treated? “The system must understand the query, find relevant agricultural information, generate an answer, and provide evidence. Specify which Transformer-based technique/model you would use for semantic understanding, document retrieval, answer generation, and evidence-based response generation. Explain the complete pipeline and justify why the final system is more reliable than a standalone LLM.	5	L3	13.2.1
12.A model has 110 million parameters. LoRA updates only 2.1% of the model parameters. Calculate: i. Number of parameters updated by LoRA. ii. Number of parameters remaining frozen. iii. Reduction in trainable parameters compared with full fine-tuning. iv. If storing each trainable parameter requires 4 bytes, calculate the memory required for the trainable parameters under: ○ Full fine-tuning ○ LoRA	5	L3	13.2.1
13. Explain why fine-tuning is required for foundation models such as BERT, GPT, and T5 after pre-training. Compare full fine-tuning with feature extraction in terms of accuracy, computational cost, and memory requirements.	5	L2	2.1.2
14. Explain how LoRA achieves parameter-efficient adaptation. Why does LoRA require fewer trainable parameters than full fine-tuning, and what are its advantages and limitations?	5	L2	13.1.2

Chapter wise Plan

Course Code and Title: 26ECAC302 Deep Learning	
Chapter Number and Title: 5. Learning without Labels — Autoencoders and Self-Supervised Learning	Planned Hours: 07

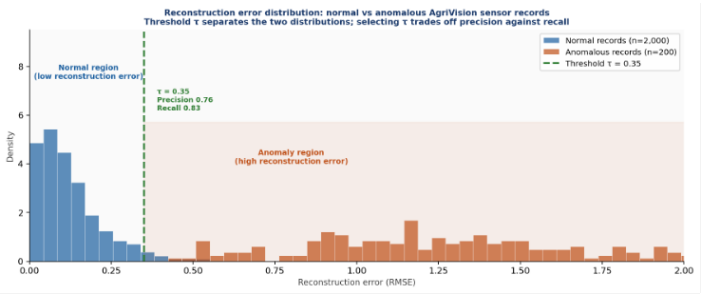
Learning Outcomes:

At the end of the topic the student should be able to:

Sr.No	TLO's	CO's	BL	CA Code
1.	Explain text as data, its inductive bias, and the limitations of sequential models for text processing.	3	L3	1.4
2.	Analyse attention, self-attention, and multi-head attention mechanisms for capturing contextual relationships in text.	3	L3	2.1
3.	Analyse the Transformer architecture and its role in modern language models.	3	L3	2.1
4.	Evaluate pre-training, foundation models, fine-tuning, adaptation, and efficient transfer techniques for NLP tasks.	3	L3	13.1
5.	Assess Transformers beyond text, hallucination, grounding, responsible use, and their application across NLP contexts.	3	L3	13.1

Lesson Schedule Class No. - Portion covered per hour		Hours
1.	The Inductive Bias for Unlabelled Data	2
2.	Limits of Supervised Learning	1
3.	Autoencoder Architecture	1
4.	Latent Space and Applications	1
5.	Self-Supervised Learning	1
6.	Application Contexts, Limitations, and Bridge	1

Review Questions

Sr.No. - Questions	TLO	BL	PI Code
1. Explain the architecture of an autoencoder and describe the roles of the encoder, latent representation, decoder, and reconstruction loss in learning from unlabelled data.	1	L3	1.4.4
2. Illustrate how an autoencoder learns a compressed latent representation from unlabelled data and analyze the effect of changing the dimensionality of the latent space on reconstruction quality.	1	L3	2.1.2
3. BERT masks tokens to create a pretext task. An autoencoder reconstructs its input. What property do these two pretext tasks share that makes them effective for representation learning?	2	L3	2.1.2
4. A student builds an overcomplete autoencoder (d_z = 420) without regularisation and achieves zero reconstruction loss. Why does anomaly detection then fail completely?	2	L3	1.4.4
5. Compare and contrast PCA, t-SNE and UMAP	3	L3	2.1.2
6. Answer the following: a. Why is the ROC curve less appropriate than the Precision-Recall curve for evaluating anomaly detection on a 99.8% normal dataset? b. A student uses Euclidean distance from the mean latent vector as the anomaly score instead of reconstruction error. When would these agree? When might they differ?	4	L3	13.1.2
7. Apply an autoencoder-based approach to a suitable application such as image denoising or anomaly detection. Explain the model architecture, training procedure, and use of the learned latent representation.	5	L3	2.1.2
8. Given the figure of “Reconstruction error distribution for AgriVision's sensor records.” Explain the figure in details 	5	L3	1.4.4

Chapter wise Plan

Course Code and Title: 26ECAC302 Deep Learning	
Chapter Number and Title: 6 Generative Deep Learning — From Prediction to Synthesis	Planned Hours:7

Learning Outcomes:

At the end of the topic the student should be able to:

Sr.No	TLO's	CO's	BL	CA Code
1.	Explain the fundamental shift from discriminative prediction to generative synthesis.	4	L2	1.4
2.	Explain and apply the principles of Variational Autoencoders, including latent space, KL divergence, ELBO and the reparameterisation trick.	4	L3	13.1
3.	Explain and analyze GAN architecture, adversarial training, minimax objective, mode collapse and GAN variants such as DCGAN, WGAN and CycleGAN.	4	L4	2.1
4.	Explain and analyze diffusion models, including forward diffusion, reverse diffusion, noise schedules, U-Net denoising and latent diffusion.	4	L4	1.4
5.	Evaluate generative models using fidelity, diversity and FID, and compare VAE, GAN and diffusion models for a given application.	4	L5	13.1
6.	Apply and evaluate generative models for data augmentation and other real-world applications while considering privacy, misuse and responsible release.	4	L5	3.3

Lesson Schedule Class No. - Portion covered per hour	Hours
1.The Inductive Bias for Generation	1
2.Why Generative Deep Learning?	1
3 Variational Autoencoders	1
4 Generative Adversarial Networks	1
5 Diffusion Models – Fundamentals	1
6. Diffusion Models – Advanced Concepts	1
7 Applications and Responsible Use	1

Review Questions

Sr.No. - Questions	TLO	BL	PI Code
1. AgriVision has a set of leaf photographs and wants to classify each photograph as healthy or diseased. Apply the concept of discriminative modelling to identify what distribution the model should learn and what output it should produce.	1	L3	13.1.1
2. A diffusion model corrupts a clean leaf image using T=1000T=1000 steps. Apply the forward diffusion equation to determine how the signal and noise contributions change as tt increases, and interpret the resulting image at a large timestep.	4	L3	1.4.4
3. Apply the concept of latent diffusion to reduce the computational cost of generating high-resolution images using a diffusion model.	4	L3	1.4.4
4. Compare VAE, GAN and diffusion models in terms of quality, diversity, stability and computational cost.	2	L2	13.1.2
5. Discuss the responsible use and misuse risks associated with generative AI.	6	L2	13.1.1
6. AgriVision needs to translate healthy leaf images into diseased leaf images, but paired healthy–diseased photographs are unavailable. Apply the CycleGAN architecture to design a suitable image-to-image translation approach and explain the role of cycle consistency.	3	L3	1.4.4
7. A diffusion model produces poor-quality images at early training stages because the denoising network cannot effectively identify image structures at different resolutions. Analyze how a U-Net architecture can be applied to address this problem.	4	L3	2.1.2
8. Analyze the roles of reconstruction loss and KL divergence in the VAE objective to determine why the generated images are blurry, and recommend a suitable modification to improve reconstruction quality.	4	L3	13.1.2
9. Design an evaluation strategy for synthetic disease images using fidelity, diversity and FID.	5	L3	13.1.2
10. A generative model produces visually realistic images but generates almost identical samples. Select appropriate evaluation measures to identify the problem and explain how fidelity and diversity should be assessed.	5	L3	3.3.2

Chapter wise Plan

Course Code and Title: 26ECAC302 Deep Learning	
Chapter Number and Title: 7. Relational and Multimodal Data — GNNs and Beyond	Planned Hours:10

Learning Outcomes:

At the end of the topic the student should be able to:

Sr.No	TLO's	CO's	BL	CA Code
1.	Explain the inductive bias of permutation invariance for graph-structured data, and the limitations of MLPs, CNNs, and RNNs on such data.	4	L2	1.4
2.	Compare GCN and GAT message-passing mechanisms, contrasting equal-weight aggregation with learned attention weighting.	4	L4	13.1
3.	Select an appropriate multimodal fusion strategy — early, late, or cross-modal — for a given data combination and data volume.	4	L3	1.4
4.	Analyze whether a problem qualifies as a sequential decision-making problem, using a structured checklist before applying Deep RL.	5	L3	2.1
5.	Describe the role of reinforcement learning (including RLHF/RLAIF) in training agentic AI systems to plan, use tools, and complete multi-step tasks	4	L2	1.4
6.	Design real-world production systems built on GNNs, multimodal fusion, or RL/agentic architectures, and justify the architecture fit.	5	L3	3.3

Lesson Schedule Class No. - Portion covered per hour	Hours
1. Introduction, Agri Vision Thread, Brief History of Architecture Families, Prerequisites,	1
2. The Inductive Bias for Relational Data, Graph-Structured Data (nodes, edges, adjacency matrix, node features, graph tasks)	1
3. Graph Neural Networks: message passing, GNN architecture, GCN	1
4. Graph Attention Networks (GAT), limitations, GNN training loop	1
5. Multimodal Deep Learning: fusion strategies, CLIP	1
6. Deep Reinforcement Learning: RL framework, exploration-exploitation, when to use DRL	1
7. RL in Agentic AI: mapping RL to agents, delayed reward, RLHF/RLAIF	1

8. Tool use, multi-agent systems, reward hacking and safety	1
9. Policy optimisation (REINFORCE to PPO), reward shaping, offline RL, evaluation	1
10.Sim-to-real gap, AgriVision's full pipeline	1

Review Questions

Sr.No. - Questions	TLO	BL	PI Code
1. Explain why permutation invariance is an important inductive bias for graph-structured data. Analyze what could happen to the predictions of an MLP if the rows representing Farm 45 and Farm 51 are exchanged.	1	L2	1.4.4
2. A 150-farm network contains proximity, water-sharing, and road relationships. Analyze why an MLP, CNN, or RNN may fail to exploit these relationships and summarize how a GNN addresses the problem through message passing.	1	L2	1.4.4
3. Compare GCN and GAT for predicting disease risk in connected farms. Explain how GCN performs neighbour aggregation and how GAT learns different attention weights for different neighbours.	2	L3	1.4.4
4. Farm 45 has three neighbours: one sharing a drainage canal, one 200 m away, and one connected through a road. Evaluate whether GCN or GAT is more appropriate when these neighbours have different levels of disease-transmission importance. Justify your answer.	2	L3	2.1.2
5. Explain the differences among early fusion, late fusion, and cross-modal attention. For a system using leaf photographs, sensor readings, and field notes, select a suitable fusion strategy and justify your choice.	3	L3	13.1.1
6. AgriVision has only 2,400 photograph–field-note pairs but has many photograph-only and text-only samples. Evaluate which multimodal fusion strategy would be appropriate and propose a suitable training approach.	3	L3	13.1.1
7. Explain the difference between a prediction problem and a sequential decision-making problem. Analyze whether selecting farms for weekly visits by an extension officer can be formulated as a Deep RL problem.	4	L3	1.4.4
8. An agricultural scheduling system must decide which farm to visit, receive information from that visit, and then select the next farm. Evaluate the problem using the four DRL suitability questions: sequential nature, reward signal, state-space size, and justification for DRL complexity.	4	L3	1.4.4

9. Explain the roles of the agent, environment, state, action, and reward in an RL-based agricultural monitoring system. Summarize how delayed rewards make RL different from supervised learning.	5	L2	1.4.4
10. Explain how RL, RLHF, and RLAIF can contribute to training an agentic AI system that plans tasks, uses tools, and completes multi-step activities.	5	L2	1.4.4
11. Demonstrate how you would design an intelligent agricultural disease-monitoring system using a GNN, multimodal fusion, or RL. Justify why the selected architecture is appropriate for the data and task.	6	L3	3.3.2
12. A smart-farming system receives leaf images, sensor time-series, field notes, and information about relationships among farms. Design a production architecture combining multimodal learning and GNNs, and justify the role of each component.	6	L3	13.1.1

Model Question Paper for Minor Examination (ISA-1)	
Course Code: 26ECAC302	Course Title: Deep Learning
Duration: 1H 15M	Max. Marks: 40
Note: Answer any two full questions	

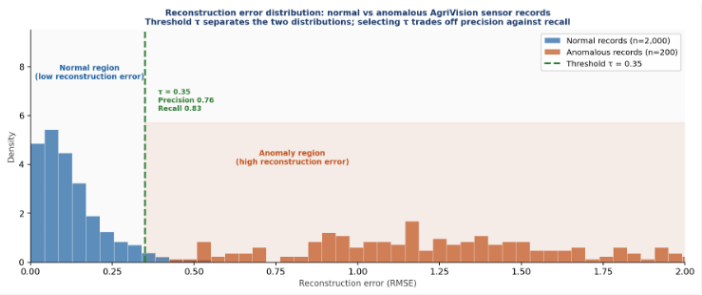
<table>
<tr><th>Q.No</th><th>Questions</th><th>Marks</th><th>CO</th><th>BL</th><th>PI Code</th></tr>
<tr><td>1a</td><td>From Table below, the unregularised MLP (RMSE 241) is worse than the regularised MLP (RMSE 219), which is worse than gradient boosting (RMSE 198). Explain why each gap exists. What does the first reveal about regularisation? The second about the suitability of deep learning for this task?

<table>
<tr><td>Linear regression</td><td>—</td><td>312</td><td>258</td><td>< 1 s</td></tr>
<tr><td>Decision tree</td><td>max_depth = 8</td><td>286</td><td>231</td><td>< 1 s</td></tr>
<tr><td>Gradient boosting</td><td>200 trees, depth 4</td><td>198</td><td>164</td><td>3 s</td></tr>
<tr><td>MLP (no regularisation)</td><td>7→64→32→1, ReLU</td><td>241</td><td>196</td><td>45 s</td></tr>
<tr><td>MLP (dropout + weight decay)</td><td>7→64→32→1, p=0.3, λ=10⁻³</td><td>219</td><td>178</td><td>45 s</td></tr>
</table></td><td>10</td><td>1</td><td>L3</td><td>1.4.4</td></tr>
<tr><td>1b</td><td>An input feature map of size 128x128x64 is processed by: (i) a regular 3x3 convolution with 128 filters and (ii) a depthwise 3x3 convolution followed by a 1x1 pointwise convolution with 128 filters. Calculate the parameters for both cases and the percentage reduction obtained using depthwise separable convolution. Ignore biases.</td><td>10</td><td>2</td><td>L3</td><td>13.2.1</td></tr>
<tr><td>2a</td><td>Given the following 3 × 3 input image matrix:
I = [10 10 10]
[10 50 50]
[10 50 50]
Using the Sobel operators:
G_x = [−1 0 1]
[−2 0 2]
[−1 0 1]
G_Y = [−1 −2 −1]
[0 0 0]
[1 2 1]
Calculate the horizontal gradient G_x, vertical gradient G_Y, and gradient magnitude at the centre pixel.Gradient magnitude: $|G| = \sqrt{(G_x^2 + G_Y^2)}$</td><td>10</td><td>2</td><td>L3</td><td>1.4.4</td></tr>
<tr><td>2b</td><td>A sequence of length T = 20 is processed using:
• A TCN with 3 layers, kernel size 3, and 32 channels.</td><td>10</td><td>1</td><td>L3</td><td>13.1.1</td></tr>
</table>

	• An LSTM with hidden size H = 32. Apply the computational complexity formulas to determine which model requires fewer operations for one forward pass. Justify your answer.				
3a	A hospital develops a system to monitor patients' heart rate continuously in real time. (a) Select the most suitable model among RNN, LSTM, and TCN. (b) Justify your selection based on sequence length and latency. (c) Identify one limitation of the chosen model.	10	2	L3	2.1.2
3b	An AgriVision image is captured at 4,032 × 3,024 pixels and resized to 224 × 224 pixels. Calculate the number of pixels before and after resizing, and explain why resizing is necessary for CNN-based image analysis.	10	1	L3	13.1.1

Model Question Paper for Minor Examination (ISA-2)	
Course Code: 26ECAC302	Course Title: Deep Learning
Duration:1H 15M	Max. Marks:40

Note: Answer any two full questions

Q.No	Questions	Marks	CO	BL	PI Code
1a	Consider the sentence: “The farmer who visited the research center last month finally planted tomatoes.” An NLP system is processing the word “planted.” Using the concept of the Attention Mechanism, explain how the model can directly identify “farmer” as an important word without passing information sequentially through all the intermediate words. Explain the roles of Query, Key, Value, Softmax, and weighted sum in this process	10	2	L3	2.1.2
1b	Given the figure of “Reconstruction error distribution for AgriVision's sensor records.” Explain the figure in details 	10	3	L3	1.4.4
2a	Answer the following: i. Why is the ROC curve less appropriate than the Precision-Recall curve for evaluating anomaly detection on a 99.8% normal dataset? ii. A student uses Euclidean distance from the mean latent vector as the anomaly score instead of reconstruction error. When would these agree? When might they differ?	10	3	L3	13.1.2
2b	A diffusion model produces poor-quality images at early training stages because the denoising network cannot effectively identify image structures at different resolutions. **Analyze** how a U-Net architecture can be applied to address this problem.	10	3	L2	2.1.2
3a	A generative model produces visually realistic images but generates almost identical samples. Select appropriate evaluation measures to identify the problem and explain how fidelity and diversity should be assessed.	10	3	L2	3.3.2

3b	An agricultural organization wants to develop three separate systems: ● System A: Classifies farmer complaints into categories such as disease, irrigation, fertilizer, and pest. ● System B: Acts as a conversational agricultural chatbot. ● System C: Converts an English agricultural advisory into Kannada. Choose the most suitable model among BERT, GPT, and T5 for each system. Justify your choices based on their architecture and capabilities.	10	3	L3	13.1.1

Duration: 1H 15M		Max. Marks: 40

Note: Answer any two full questions from each unit-I and unit-II and one full question from unit-III

UNIT 1

Q.No	Questions	Marks	CO	BL	PI Code
1a	A smart manufacturing unit has 600 labelled thermal images of electrical panels (420 normal, 180 faulty) and requires an automated fault-detection model to run on an edge device with less than 150 ms inference time per image. (a) Using a Deep Learning model-selection decision framework, evaluate the suitability of Deep Learning considering dataset size, class imbalance, computational constraints, and image characteristics. Recommend an appropriate training strategy and architecture family. (b) Design an evaluation and deployment strategy. Justify suitable performance metrics beyond accuracy, and explain how you would address overfitting, class imbalance, and the inference-time constraint. Briefly state when a traditional ML model using hand-crafted features might be preferable	10	1	L3	1.4.4
1b	A CNN takes a 32 × 32 × 3 input image and applies a 3 × 3 convolution with 16 filters, stride 1, and padding 1. Calculate the output tensor dimensions and total number of trainable parameters, including one bias for each filter.	10	2	L3	13.2.1
2a	Given the following 3 × 3 input image matrix: I = [10 10 10] [10 50 50] [10 50 50] Using the Sobel operators: G_x = [−1 0 1] [−2 0 2] [−1 0 1] G_Y = [−1 −2 −1] [0 0 0] [1 2 1] Calculate the horizontal gradient G_x, vertical gradient G_Y, and gradient magnitude at the centre pixel.Gradient magnitude: $\|G\| = \sqrt{(G_x^2 + G_Y^2)}$	10	2	L3	1.4.4
2b	A farmer's monitoring system stores insect counts from the previous 45 days as independent input features to an MLP. What are two limitations of this fixed-window representation compared to an LSTM?	10	2	L3	13.1.1
3a	A hospital develops a system to monitor patients' heart rate continuously in real time. (a) Select the most suitable model among RNN, LSTM, and TCN.	10	1	L3	2.1.2

	(b) Justify your selection based on sequence length and latency. (c) Identify one limitation of the chosen model. Chap 3				
3b	A YOLO model detects three lesions of the same disease with confidence scores 0.92, 0.78, and 0.65. Two bounding boxes have IoU = 0.68, while the third has IoU = 0.20 with the highest-confidence box. (a) Which box is selected first during NMS? (b) Which box is suppressed using an IoU threshold of 0.5? (c) Explain how NMS helps produce the final detection result. Chap 2	10	2	L3	13.1.1
	UNIT 2				
4a	An agricultural advisory system receives the sentence:"The experienced farmer carefully sprayed the pesticide on the infected crop yesterday." The system needs to understand several relationships simultaneously: farmer → sprayed, pesticide → sprayed, infected → crop, and yesterday → sprayed. Would single self-attention or multi-head attention be more appropriate for this task? Justify your choice and explain how different attention heads can capture different semantic or syntactic relationships simultaneously.	10	2	L3	2.1.2
4b	Answer the following: i. Why is the ROC curve less appropriate than the Precision-Recall curve for evaluating anomaly detection on a 99.8% normal dataset? ii. A student uses Euclidean distance from the mean latent vector as the anomaly score instead of reconstruction error. When would these agree? When might they differ?	10	3	L3	1.4.4
5a	Apply an autoencoder-based approach to a suitable application such as image denoising or anomaly detection. Explain the model architecture, training procedure, and use of the learned latent representation.	10	3	L3	13.1.2
5b	Design an evaluation strategy for synthetic disease images using fidelity, diversity and FID.	10	3	L2	2.1.2
6a	A diffusion model produces high-quality images but is too slow to deploy. Suggest approaches based on the chapter to reduce inference cost	10	3	L2	3.3.2
6b	An agricultural organization wants to develop three separate systems: • System A: Classifies farmer complaints into categories such as disease, irrigation, fertilizer, and pest. • System B: Acts as a conversational agricultural chatbot.	10	3	L3	13.1.1

	• System C: Converts an English agricultural advisory into Kannada. Choose the most suitable model among BERT, GPT, and T5 for each system. Justify your choices based on their architecture and capabilities.				
	UNIT 3				
7a	A 150-farm network contains proximity, water-sharing, and road relationships. Analyze why an MLP, CNN, or RNN may fail to exploit these relationships and summarize how a GNN addresses the problem through message passing.	10	4	L3	2.1.2
7b	Farm 45 has three neighbours: one sharing a drainage canal, one 200 m away, and one connected through a road. Evaluate whether GCN or GAT is more appropriate when these neighbours have different levels of disease-transmission importance. Justify your answer.	10	4	L3	1.4.4
8a	A smart-farming system receives leaf images, sensor time-series, field notes, and information about relationships among farms. Design a production architecture combining multimodal learning and GNNs, and justify the role of each component.	10	4	L3	3.3.2
8b	Demonstrate how you would design an intelligent agricultural disease-monitoring system using a GNN, multimodal fusion, or RL. Justify why the selected architecture is appropriate for the data and task.	10	4	L3	13.1.1